

Task 2: Installing IIS Using Custom Data & Custom Script Extension

◆ Step 1: Create Windows VM with Custom Data

- Windows Virtual Machine created
 - PowerShell commands added under **Custom Data**
 - Custom Data executes **only once during VM provisioning**
-

◆ Step 2: Create Storage Account and Upload Script

Steps:

1. Storage Account created
2. Blob container created
3. PowerShell script uploaded

Script automates:

- IIS installation
 - Google Chrome installation
-

◆ Step 3: Execute Script Using Custom Script Extension (CSE)

PowerShell Script Used

```
param (  
    [bool]$InstallIIS = $true,
```

```

        [bool]$InstallChrome = $true,
        [string]$IISMessage = "Hello from IIS"
    )

    $OS = (Get-CimInstance Win32_OperatingSystem).Caption

    if ($InstallIIS) {
        if ($OS -like "*Server*") {
            Install-WindowsFeature -Name Web-Server
        -IncludeManagementTools
        } else {
            Enable-WindowsOptionalFeature -Online -FeatureName
IIS-WebServerRole -All -NoRestart
        }

        Start-Service W3SVC
        Set-Service W3SVC -StartupType Automatic

        $html = @"
<html>
<body>
<h1>$IISMessage</h1>
<p>IIS is running successfully</p>
</body>
</html>
"@

        $html | Out-File "C:\inetpub\wwwroot\index.html" -Encoding UTF8
    }

    if ($InstallChrome) {
        $ChromeInstaller = "$env:TEMP\chrome_installer.exe"
        $ChromeURL =
"https://dl.google.com/chrome/install/latest/chrome_installer.exe"

        Invoke-WebRequest -Uri $ChromeURL -OutFile $ChromeInstaller
        Start-Process -FilePath $ChromeInstaller -ArgumentList "/silent
/install" -Wait
        Remove-Item $ChromeInstaller -Force
    }

```

```
exit 0
```

Script Execution Location

```
C:\WindowsAzure\Logs\Plugins\Microsoft.Compute.CustomScriptExtension  
\<Version>\Downloads\<n>\
```

```
![CSE Logs](./Images/Day-6/Screenshot 2026-02-09 174638.png)
```

◆ Step 4: Verify Custom Data Storage

Custom Data is stored in Base64 format at:

```
C:\AzureData\CustomData.bin
```

```
![Custom Data File](./Images/Day-6/Screenshot 2026-02-09 174550.png)
```

◆ Step 5: Access IIS Using Public IP

Access URL:

```
http://<VM-PUBLIC-IP>
```

Result:

- IIS homepage loads
- Displays custom message defined in script

```
![IIS Output](./Images/Day-6/Screenshot 2026-02-09 174935.png)
```

Final Outcome

- ✓ VM Scale Set created from custom image
- ✓ Autoscaling triggered and validated
- ✓ IIS installed automatically

- ✓ Custom Script Extension executed successfully
- ✓ Application accessible via Public IP

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CreateVm-microsoftwindowsdesktop.windows-11-win11-20260209094609 | Overview

Deployment

Search

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✓ Your deployment is complete

Deployment name: CreateVm-microsoftwindowsdesktop.windows-11-win11-20260209094609 Start time: 2/9/2026, 9:54:28 AM
Subscription: Vanguard Class Training Correlation ID: 347c1779-f58-4d79-8e72-842594d9085
Resource group: syedfuzail2009-rg

Deployment details

Next steps

- [Setup auto-shutdown](#) Recommended
- [Monitor VM health, performance and network dependencies](#) Recommended
- [Run a script inside the virtual machine](#) Recommended

[Go to resource](#) [Create another VM](#)

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CreateVmss-0.0.1-20260209124534 | Overview

Deployment

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✓ Your deployment is complete

Deployment name: CreateVmss-0.0.1-20260209124534 Start time: 2/9/2026, 12:51:41 PM
Subscription: Vanguard Class Training Correlation ID: 2c2ae050-7986-4707-ac87-b77b47251e81
Resource group: syedfuzail2009-rg

Deployment details

Next steps

[Go to resource](#)

Deployment succeeded
Deployment 'CreateVmss-0.0.1-20260209124534' to resource group 'syedfuzail2009-rg' was successful.
[Go to resource](#) [Go to resource group](#)

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